ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C4 · ENERGY

Forms of energy

★☆☆
Grade 5

Learning goals

- Identify forms of energy.
- Complete energy chains.

1 **Sort the examples by the main form of energy.** [6]

SORT THESE

a spinning fan **a moving bus** **a plate of rice** **a drum beat** **a battery** **a firefly**

Kinetic	Chemical (stored)	Light or sound
a moving bus	a battery	a firefly
a spinning fan	a plate of rice	a drum beat

2 **Complete the energy chains.** [8]

Device	Energy in	Useful energy out	Wasted energy
electric fan	electrical	kinetic (moving air)	heat, sound
solar panel	light	electrical	heat
radio	electrical	sound	heat
gas stove	chemical	heat (thermal)	light, heat to the room

3 **Fill in the gaps.** [5]

WORD BANK

created **colder** **hotter** **insulators** **conductors** **rises** **falls**

Energy cannot be created or destroyed. Heat moves from hotter objects to colder ones. When an object gains heat its temperature rises. Metals are good thermal conductors.

ANSWER KEY

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THINK IT THROUGH · C4 · ENERGY

Following the energy

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Grade 5

Learning goals

- Explain energy transfers.
- Explain heating and cooling.

1 **Describe the energy changes when a rickshaw puller rides up a bridge and then freewheels down.** [3]

Chemical energy from food → kinetic energy of the rickshaw; going up, kinetic energy changes into (gravitational) potential energy; coming down, potential energy changes back into kinetic energy. Some energy is wasted as heat and sound.

2 **"When a ball stops bouncing, its energy has been destroyed." Do you agree? Explain.** [2]

No: energy cannot be destroyed; it has been transferred to the surroundings as heat and sound each time the ball hits the ground.

3 **Explain why a metal spoon in hot tea gets hot quickly, but a wooden spoon does not.** [2]

Metal is a thermal conductor, so heat moves through it easily; wood is an insulator, so heat moves slowly.

4 **A cold bottle of water and a hot cup of tea are left on a table. What happens to their temperatures? Explain using the word *heat*.** [2]

The tea loses heat to the room and cools down; the cold water gains heat from the room and warms up; both end at room temperature.

5 **Why is an LED bulb more efficient than a filament bulb?** [1]

It changes more of its electrical energy into light and wastes less as heat.

ANSWER KEY

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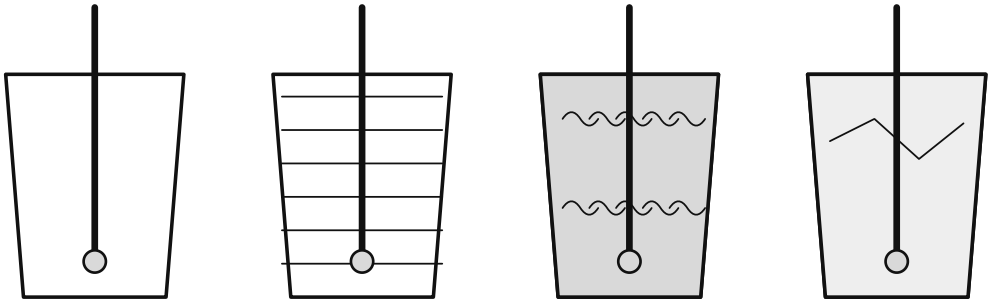
INVESTIGATE · C4 · ENERGY

Which wrapping keeps tea hottest?

★★☆
Grade 5

Learning goals

- Plan a fair test.
- Plot several lines on a graph.
- Explain results using insulation.



no cover **newspaper** **wool** **foil**

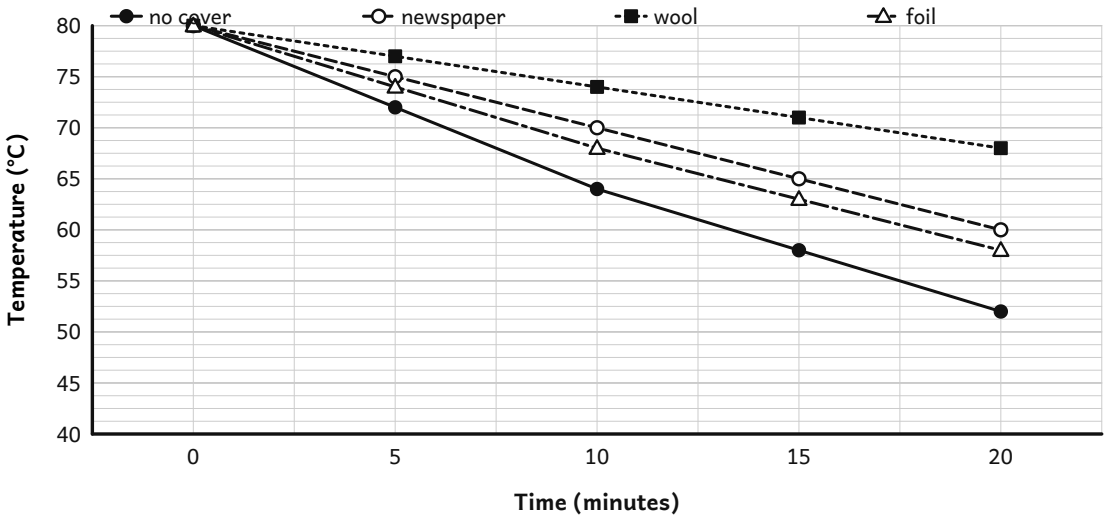
Class 5D poured 200 ml of water at 80 °C into four identical cups, wrapped three of them in different materials, and measured the temperature every 5 minutes.

1 **Name the variables.** [3]

Change	the wrapping material
Measure	temperature every 5 minutes (°C)
Keep the same	volume and starting temperature of water, type of cup, room, time, thermometer

2 **Here are the results. Plot all four on one line graph (use a different line or symbol for each).** [4]

Time (min)	0	5	10	15	20
no cover	80	72	64	58	52
newspaper	80	75	70	65	60
wool	80	77	74	71	68
foil	80	74	68	63	58



3 **Which material was the best insulator? Use numbers in your answer.** [2]

Wool: the water only cooled by 12 °C (80 → 68) in 20 minutes, compared with 28 °C for no cover.

4 **Why do you think wool works so well?** [1]

Wool traps lots of air, and air is a poor conductor of heat, so heat escapes slowly.

5 **Suggest a real use for what the class found out.** [1]

e.g. wrapping a tiffin box or a flask; warm woollen clothes in winter; insulating a roof.

ANSWER KEY

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REAL-WORLD CASE STUDY · C4 · ENERGY

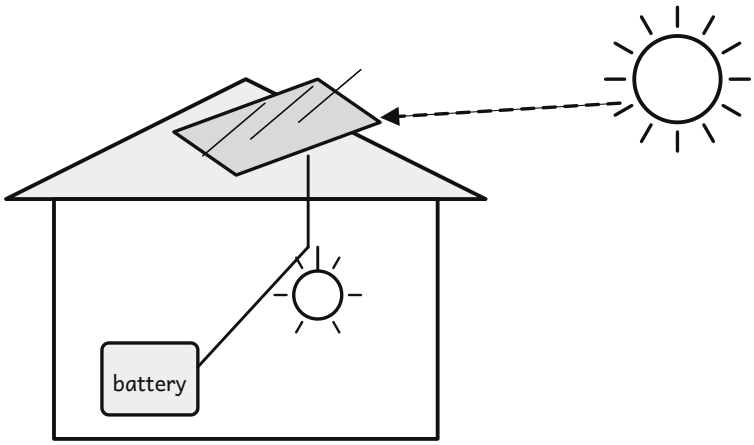
Power from the sun: solar home systems

★★★
Grade 5

Learning goals

- Trace energy transfers in a real system.
- Use maths to plan energy use.
- Weigh benefits and problems.

Science | Maths | Economics | Environment | Technology



solar panel → battery → LED light, fan, phone charger

Twenty years ago, millions of village homes in Bangladesh had no electricity. At night families used **kerosene lamps**, which give dim light, smoky air and a risk of fire. Since the early 2000s, millions of homes have bought **solar home systems**, making Bangladesh one of the world's biggest users of small solar systems.

A solar panel on the roof turns sunlight into electricity during the day. This charges a **battery**. At night the battery powers **LED lights**, a small **fan**, a **phone charger** and sometimes a TV. Children can study after dark, and shops can stay open longer.

Problems include the **cost** of buying a system, batteries that wear out after a few years and must be **recycled** safely (old batteries contain harmful lead or chemicals), and less power on cloudy monsoon days.

Based on reports about Bangladesh's solar home system programme (IDCOL, World Bank).

1 **Write the energy chain for an LED light powered by a solar home system at night.** [3]

light (Sun) → electrical (panel) → chemical (stored in battery) → electrical → light (+ heat) from the LED

2 **Energy maths. A 50 W panel gets 5 hours of good sunshine a day.** [4]

a) How many watt-hours (Wh) of energy can it make in a day? (W × hours)

50 × 5 = 250 Wh

b) A family uses 3 LED bulbs of 5 W for 4 hours. How much energy is that?

3 × 5 × 4 = 60 Wh

c) They also use a 20 W fan for 6 hours. Total energy used?

20 × 6 = 120 Wh; total 60 + 120 = 180 Wh

d) Is there enough energy left to charge a phone (10 Wh)?

Yes: 250 − 180 = 70 Wh left, more than 10 Wh.

3 **Give two ways solar lights are better than kerosene lamps.** [2]

No smoke, so healthier lungs; brighter light for studying; no fire risk; no fuel to buy; no carbon dioxide while working.

4 **Old batteries contain harmful materials. Suggest a way to deal with them safely.** [1]

Return them to the seller or a recycling centre so the materials are recovered and do not pollute soil and water.

ANSWER KEY

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REVIEW AND CHALLENGE · C4 · ENERGY

Show what you know

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Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

L	I	G	H	T	L	D	A	V	T	W
U	W	E	M	S	T	O	R	E	D	T
K	B	A	T	T	E	R	Y	N	F	K
C	I	I	N	S	U	L	A	T	O	R
O	S	N	N	H	J	M	C	N	F	B
I	O	E	E	C	D	K	Y	I	J	J
B	U	N	D	T	C	N	P	S	S	H
W	N	E	C	C	I	K	J	T	O	E
V	D	R	C	A	S	C	H	Y	L	E
H	G	G	H	E	A	T	C	K	A	R
V	G	Y	D	R	C	P	Y	J	R	H

battery **energy** **heat** **insulator** **kinetic** **light** **solar** **sound** **stored**

2 **Which form of energy is stored in food?** [1]

- ○ kinetic
- ● **chemical**
- ○ sound
- ○ light

3 **A hot iron is left on a cold table. Which is true?** [1]

- ○ the iron gains heat
- ● **heat moves from the iron to the table**
- ○ cold moves into the iron
- ○ the temperatures stay the same

4 **Draw an energy chain for a person playing a dhol (drum).** [2]

My energy chain

chemical (food) → kinetic (moving hands and drum skin) → sound (+ heat)

5 **Challenge. "We should save energy, but energy cannot be destroyed." Explain why both statements can be true.** [3]

Energy is never destroyed, but useful energy (like fuel or electricity) gets spread out as heat that we cannot use again; saving energy means using less of the useful stored energy (fuel), which is limited and costs money.